Linux Command Line and Shell Scripting BIBLE

4th Edition

Richard Blum
Christine Bresnahan

WILEY

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Linux® Command Line and Shell Scripting BIBLE

Richard Blum

Christine Bresnahan

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Published simultaneously in Canada and the United Kingdom

ISBN: 978-1-119-70091-3
ISBN: 978-1-119-70094-4 (ebk)
ISBN: 978-1-119-70093-7 (ebk)

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Library of Congress Control Number: 2020949805

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To the Lord God Almighty, "in whom are hidden all the treasures of wisdom and knowledge."
— Colossians 2:3

About the Authors

Richard Blum has worked in the IT industry for more than 30 years as both a systems and a network administrator. During that time, he's had the opportunity to work with lots of different computer products, including Windows, NetWare, Cisco, Avaya, different flavors of UNIX, and of course, Linux. Over the years he's also volunteered for several nonprofit organizations to help support small networks that had little financial support. Rich is the author of many Linux-based books for total Linux geeks and teaches online courses in Linux and web programming. When he's not busy being a computer nerd, Rich enjoys playing piano and bass guitar and spending time with his wife, Barbara, and their two daughters, Katie Jane and Jessica.

Christine Bresnahan started working with computers more than 30 years ago in the IT industry as a systems administrator. Christine is an adjunct professor at Ivy Tech Community College, where she teaches Linux certification and Python programming classes. She also writes books and produces instructional resources for the classroom. During her downtime, Christine enjoys spending time with her husband and family, hiking, and gardening.

About the Technical Editor

Jason W. Eckert is an experienced technical trainer, consultant, and best-selling author in the technology industry. With 45 industry certifications, 25 published textbooks, and over 30 years of technology and programming experience, Jason brings his expertise to every class that he teaches at triOS College. For more information about him, visit `jasoneckert.net`.

Acknowledgments

First, all glory and praise go to God, who through His Son, Jesus Christ, makes all things possible and gives us the gift of eternal life.

Many thanks go to the fantastic team of people at John Wiley & Sons for their outstanding work on this project. Thanks to Kenyon Brown, the acquisitions editor, for offering us the opportunity to work on this book. Also thanks to Patrick Walsh, the project editor, for keeping things on track and making this book more presentable. Thanks, Pat, for all your hard work and diligence. The technical editor, Jason Eckert, did a wonderful job of double-checking all the work in the book, plus making suggestions to improve the content. Thanks to Saravanan Dakshinamurthy and his team for their endless patience and diligence to make our work readable. We would also like to thank Carole Jelen at Waterside Productions, Inc., for arranging this opportunity for us, and for helping us out in our writing careers.

Christine would like to thank her husband, Timothy, for his encouragement, patience, and willingness to listen, even when he has no idea what she is talking about. Rich would like to thank his wife, Barbara, for the life-sustaining baked goods she readily prepared to help him keep up his energy while writing!

Introduction

Welcome to the fourth edition of *Linux Command Line and Shell Scripting Bible*. Like all books in the *Bible* series, you can expect to find both hands-on tutorials and real-world information, as well as reference and background information that provides a context for what you are learning. This book is a fairly comprehensive resource on the Linux command line and shell commands. By the time you have completed *Linux Command Line and Shell Scripting Bible,* you will be well prepared to write your own shell scripts that can automate practically any task on your Linux system.

Who Should Read This Book

If you're a systems administrator in a Linux environment, you'll benefit greatly by knowing how to write shell scripts. The book doesn't walk you through the process of setting up a Linux system, but after you have it running, you'll want to start automating some of the routine administrative tasks. That's where shell scripting comes in, and that's where this book helps you out. This book demonstrates how to automate any administrative task using shell scripts, from monitoring system statistics and data files to generating reports for your boss.

If you're a home Linux enthusiast, you'll also benefit from *Linux Command Line and Shell Scripting Bible*. Nowadays, it's easy to get lost in the graphical world of prebuilt widgets. Most desktop Linux distributions try their best to hide the Linux system from the typical user. However, sometimes you must know what's going on under the hood. This book shows you how to access the Linux command-line prompt and what to do when you get there. Often, performing simple tasks, such as file management, can be done more quickly from the command line than from a fancy graphical interface. You can use a wealth of commands from the command line, and this book shows you how to use them.

How This Book Is Organized

This book leads you through the basics of the Linux command line and into more complicated topics, such as creating your own shell scripts. The book is divided into four parts, each one building on the previous parts.

Part I assumes that you either have a Linux system running or are looking into getting a Linux system. Chapter 1, “Starting with Linux Shells,” describes the parts of a total Linux system and shows how the shell fits in. After describing the basics of the Linux system, this part continues with the following:

- Using a terminal emulation package to access the shell (Chapter 2)
- Introducing the basic shell commands (Chapter 3)
- Using more advanced shell commands to peek at system information (Chapter 4)
- Understanding what the shell is used for (Chapter 5)
- Working with shell variables to manipulate data (Chapter 6)
- Understanding the Linux filesystem and security (Chapter 7)
- Working with Linux filesystems from the command line (Chapter 8)
- Installing and updating software from the command line (Chapter 9)
- Using the Linux editors to start writing shell scripts (Chapter 10)

In Part II, you begin writing shell scripts. As you go through the chapters, you'll do the following:

- Learn how to create and run shell scripts (Chapter 11)
- Alter the program flow in a shell script (Chapter 12)
- Iterate through code sections (Chapter 13)
- Handle data from the user in your scripts (Chapter 14)
- See different methods for storing and displaying data from your script (Chapter 15)
- Control how and when your shell scripts run on the system (Chapter 16)

Part III dives into more advanced areas of shell script programming, including these things:

- Creating your own functions to use in all your scripts (Chapter 17)
- Utilizing the Linux graphical desktop for interacting with your script users (Chapter 18)
- Using advanced Linux commands to filter and parse data files (Chapter 19)
- Using regular expressions to define data (Chapter 20)
- Learning advanced methods of manipulating data in your scripts (Chapter 21)
- Working with advanced features of scripting to generate reports from raw data (Chapter 22)
- Modifying your shell scripts to run in other Linux shells (Chapter 23)

The last section of the book, Part IV, demonstrates how to use shell scripts in real-world environments. In this part, you will learn these things:

- How to put all the scripting features together to write your own scripts (Chapter 24)
- How to organize and track your script versions using the popular git software (Chapter 25)

Conventions and features

You will find many different organizational and typographical features throughout this book designed to help you get the most out of the information.

Tips and warnings

Whenever the authors want to bring something important to your attention, the information appears in a Warning.

> **WARNING**
>
> **This information is important and is set off in a separate paragraph with a special icon. Warnings provide information about things to watch out for, whether simply inconvenient or potentially hazardous to your data or systems.**

For additional items of interest that relate to the chapter text, the authors use Tip.

> **TIP**
>
> **Tips provide additional, ancillary information that is helpful, but somewhat outside of the current presentation of information.**

Minimum Requirements

Linux Command Line and Shell Scripting Bible doesn't focus on any specific Linux distribution, so you can follow along in the book using any Linux system you have available. The bulk of the book references the Bash shell, which is the default shell for most Linux systems.

Where to Go from Here

After you've finished reading *Linux Command Line and Shell Scripting Bible*, you're well on your way to incorporating Linux commands in your daily Linux work. In the ever-changing world of Linux, it's always a good idea to stay in touch with new developments. Often, Linux distributions change, adding new features and removing older ones. To keep your knowledge of Linux fresh, always stay well informed. Find a good Linux forum site and monitor what's happening in the Linux world. Many popular Linux news sites, such as Slashdot and DistroWatch, provide up-to-the-minute information about new advances in Linux.